

Briefing Document

Overland Erosion: From Rainfall Impact to Watershed Sediment Yield

Executive Summary

Overland erosion represents the critical interface between hydrology and sediment transport and completes the systemic cycle of:

Rainfall → Infiltration → Runoff → Erosion → Transport → Deposition

This document synthesizes the physical mechanisms governing soil detachment and transport, empirical and process-based models for erosion prediction, and scaling approaches used to estimate watershed sediment yield.

A central framework in modern erosion analysis is the balance between rainfall-driven detachment capacity:

$$D_c$$

and flow-driven transport capacity:

$$T_c$$

where total erosion is governed by the smaller of the two processes:

$$E = \min (D_c, T_c)$$

Although the Universal Soil Loss Equation (USLE) remains the foundational empirical model for agricultural and plot-scale applications, modern management increasingly relies on process-based approaches such as the Water Erosion Prediction Project (WEPP), which accounts for deposition, event dynamics, and sediment routing.

Understanding these processes is increasingly important because climate change may amplify rainfall intensity and runoff shear stresses, thereby increasing sediment loads affecting reservoirs and stormwater systems.

1. Physical Processes of Overland Erosion

Erosion occurs as a multistage process embedded within a larger hydrologic system:

Rainfall → Infiltration → Runoff → Erosion → Transport → Deposition

1.1 Types of Overland Erosion

Erosion commonly develops through four stages of increasing severity and channel definition.

Splash Erosion

Splash erosion occurs when raindrop impact detaches soil particles.

Detachment occurs when rainfall kinetic energy exceeds soil resistance resulting from:

- cohesion,
- adhesion.

The process is strongly dependent upon:

- rainfall intensity,
 - soil characteristics.
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Sheet Erosion

Sheet erosion involves shallow overland flow that selectively removes fine particles from hillslopes.

Over time, selective transport may produce an armor layer:

$$d_{\text{surface}} > d_{\text{subsurface}}$$

where coarse particles become concentrated at the surface.

Rill Erosion

Rill erosion develops when concentrated runoff creates small channels.

Characteristics:

- narrow flow paths,
 - concentrated high velocities,
 - generally removable through normal tillage.
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Gully Erosion

Gully erosion represents severe channel enlargement.

Typical threshold:

$$h > 300 \text{ mm}$$

Characteristics:

- deep channels,
- permanent landscape alteration,
- potential infrastructure damage.

Prevention commonly requires vegetation coverage exceeding:

$$70\%$$

1.2 Conceptual Framework: Detachment versus Transport

Modern erosion theory uses an analogy to sediment transport capacity concepts.

Two competing processes control erosion:

Detachment Capacity

Rainfall-driven detachment:

$$D_c$$

Transport Capacity

Flow-driven transport:

$$T_c$$

typically governed by shear stress:

$$\tau \approx \rho g h S$$

where:

- ρ = water density
- g = gravitational acceleration
- h = flow depth
- S = slope

The overall erosion rate is controlled by the smaller process:

$$E = \min (D_c, T_c)$$

2. Quantitative Modeling of Erosion

2.1 Rainfall and Detachment Mechanics

Soil detachment depends on raindrop characteristics.

Terminal Velocity

Raindrop terminal velocity can be approximated by:

$$V \approx 120d^{0.35}$$

where:

- V = terminal velocity
 - d = raindrop diameter
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Rainfall Kinetic Energy

Rainfall kinetic energy may be expressed as:

$$KE = 916 + 330 \log (i)$$

where:

- i = rainfall intensity
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Ellison (1947) Relationship

Detached soil mass may be estimated by:

$$G \propto V^{4.33} d^{1.07} i^{0.65}$$

where:

- G = detached soil mass
 - V = raindrop velocity
 - d = raindrop diameter
 - i = rainfall intensity
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2.2 Universal Soil Loss Equation (USLE)

USLE estimates annual soil loss:

$$E = RKLSCP$$

where:

- E = annual soil loss per unit area
 - R = rainfall erosivity factor
 - K = soil erodibility factor
 - LS = topographic factor
 - C = vegetation cover factor
 - P = conservation practice factor
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Factor	Description	Influencing Variables
<i>R</i>	Rainfall erosivity	Rainfall intensity and frequency
<i>K</i>	Soil erodibility	Soil type
<i>LS</i>	Topographic factor	Slope length and steepness
<i>C</i>	Vegetation cover	Crop type and residue management
<i>P</i>	Conservation practice	Terracing, contouring, strip cropping

Limitations of USLE

USLE:

- is empirical,
 - is largely plot-scale,
 - does not simulate deposition,
 - does not route sediment,
 - does not incorporate event dynamics.
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2.3 Modern Process-Based Models

RUSLE / RUSLE2

Revised versions of USLE commonly used for conservation planning.

Applications include:

- agriculture,
 - forests,
 - mining,
 - construction sites.
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WEPP

The Water Erosion Prediction Project model includes:

- climate processes,
- plant growth,
- hydrology,
- residue decomposition.

It computes both:

$$D_c$$

and deposition processes.

SWAT

Used for:

- long-term watershed hydrology,
 - sediment routing.
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SWMM

Applied primarily in urban environments for:

- runoff routing,
 - buildup and washoff modeling.
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3. Watershed Sediment Yield

Sediment yield represents the portion of eroded material reaching a watershed outlet.

3.1 Definitions

Gross Erosion

Total erosion from all watershed surfaces.

Sediment Yield

Sediment delivered to a control point.

Sediment Delivery Ratio (SDR)

Defined as:

$$SDR = \frac{\text{Sediment Yield}}{\text{Gross Erosion}}$$

3.2 Sediment Yield Estimation Methods

Rating Curve Method

Sediment yield may be estimated by:

$$Y = \int Q_s f(Q) dQ$$

where:

- Q_s = sediment discharge
 - $f(Q)$ = discharge probability distribution
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Reservoir Method

Sediment yield estimated using:

$$Y = \frac{A}{TE}$$

where:

- A = accumulated sediment
 - TE = trap efficiency
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USLE + SDR Method

Sediment yield:

$$Y = E_{\text{gross}} \times SDR$$

4. Urban and Construction Site Erosion

Construction activities can generate substantially higher erosion rates because of disturbed soils.

Typical practice factors include:

High Erosion Conditions

Compacted surfaces:

$$P \approx 1.30$$

Low Erosion Conditions

Sediment basins with flocculation:

$$0.10 < P < 0.20$$

Urban Buildup and Washoff

Pollutants accumulate on impervious surfaces during dry periods.

During rainfall:

- rapid initial washoff occurs,
 - followed by a decay phase.
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5. Case Study: Annual Sediment Yield Calculation

Example:

Central Illinois farm:

800 acres

Step 1: Soil Loss (USLE)

Given:

$$\begin{aligned}R &= 200 \\K &= 0.33 \\LS &= 0.697 \\C &= 0.004 \\P &= 0.5\end{aligned}$$

Calculate:

$$\begin{aligned}E &= (200)(0.33)(0.697)(0.004)(0.5) \\E &= 0.092 \text{ tons/acre/year}\end{aligned}$$

Step 2: Gross Erosion

$$\begin{aligned}E_{\text{gross}} &= 0.092 \times 800 \\&= 73.6 \text{ tons/year}\end{aligned}$$

Step 3: Sediment Delivery Ratio

For:

$$\begin{aligned}&800 \text{ acres} \\SDR &\approx 0.417\end{aligned}$$

Step 4: Sediment Yield

$$\begin{aligned}Y &= 73.6 \times 0.417 \\&= 30.7 \text{ tons/year}\end{aligned}$$

6. Engineering and Environmental Implications

6.1 Climate Change Amplification

Climate change acts as a multiplier for erosion processes.

Increased rainfall intensity raises:

$$R \uparrow$$

and increased runoff magnitudes increase:

$$\tau \uparrow$$

thus:

$$D_c \uparrow$$

and

$$T_c \uparrow$$

leading to increased erosion potential:

$$E \uparrow$$

6.2 Design and Management

Key considerations include:

Vegetation Control

Maintaining vegetation remains the most effective erosion-control measure through modification of:

- C
- P

factors.

Infrastructure Impacts

Higher sediment loads can:

- reduce reservoir storage,
- overwhelm stormwater systems.

Regulatory Implications

Design storms and erosion-control practices increasingly require updates reflecting changing rainfall intensities.

Key Takeaways

- Overland erosion links hydrology and sediment transport.
- Erosion is controlled by detachment and transport capacity.
- USLE remains foundational, while WEPP and other process-based models provide more physically based predictions.
- Sediment yield depends strongly on watershed-scale delivery processes.
- Climate change may amplify both detachment and transport mechanisms.